

It's common to hear that bitcoin supply will max out at 21 million units by 2140. This is a function of continuing to divide the units of supply released by a factor of two every four years. As of January 1, 2017, already 76.6 percent of bitcoin's supply had been brought into existence,<sup>13</sup> and by the time the next block reward halving happens in 2020, 87.5 percent of the bitcoin ever to be minted will be in existence. A few years after 2100, we will reach a supply of 20,999,999 bitcoin, which is effectively 21 million. It is bitcoin's scarce supply schedule that makes many think of it as digital gold.<sup>14</sup>

## THE BIRTH OF ALTCOINS

Within a couple years of launching, it had become clear that bitcoin was the first fully decentralized cryptocurrency to gain significant adoption, but there were some aspects with which people were not fully satisfied. For example, bitcoin's 10-minute block time meant that, depending on when a consumer hit send, it could take up to 10 minutes, sometimes more, for the transaction to be appended to Bitcoin's blockchain.

Often this delay was more of an issue for the merchant than the consumer, as the merchants needed to know they were getting paid before they could release a good or service. Others worried about bitcoin's hash function in the proof-of-work process, because hardware was being created that specialized in this hash function and would lead to increased centralization of the mining network. For a decentralized currency, increased centralization of the machines that